21ST.py - C:/Users/thala/Downloads/FODS/21ST.py (3.14.3)

File Edit Format Run Options Window Help

```python
import numpy as np
from scipy import stats

data = np.array([12.5, 13.2, 11.8, 12.9, 13.5, 12.1, 13.0, 12.7, 13.3, 12.4])

n = int(input("Enter sample size: "))
confidence = float(input("Enter confidence level (e.g. 0.95): "))
precision = float(input("Enter desired precision: "))

sample = data[:n]

mean = np.mean(sample)
std = np.std(sample, ddof=1)

alpha = 1 - confidence
t_value = stats.t.ppf(1 - alpha / 2, n - 1)

margin = t_value * (std / np.sqrt(n))

lower = mean - margin
upper = mean + margin

print("Sample Mean:", round(mean, 2))
print("Confidence Interval:", (round(lower, 2), round(upper, 2)))

if margin <= precision:
    print("Desired precision achieved")
else:
    print("Desired precision not achieved")
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
====================================================================== RESTART: C:
/Users/thala/Downloads/FODS/21ST.py ==========================================
========================
Enter sample size: 10
Enter confidence level (e.g. 0.95): 0.95
Enter desired precision: 0.5
Sample Mean: 12.74
Confidence Interval: (np.float64(12.35), np.float64(13.13))
Desired precision achieved
>>>
```

Ln: 11 Col: 0

22ND.py - C:/Users/thala/Downloads/FODS/22ND.py (3.14.3)

File Edit Format Run Options Window Help

```python
import pandas as pd
import numpy as np
from scipy import stats

# Create sample ratings
data = {
    "Rating": [4, 5, 4, 3, 5, 4, 5, 4, 3, 5]
}

df = pd.DataFrame(data)

ratings = df["Rating"]

mean = ratings.mean()
std = ratings.std()
n = len(ratings)

confidence = 0.95
alpha = 1 - confidence

t_value = stats.t.ppf(1 - alpha / 2, n - 1)
margin = t_value * (std / np.sqrt(n))

lower = mean - margin
upper = mean + margin

print("Average Rating:", round(mean, 2))
print("95% Confidence Interval:", (round(lower, 2), round(upper, 2))

if mean >= 4:
    print("Customer Satisfaction: High")
else:
    print("Customer Satisfaction: Moderate")
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==================================================================== RESTART: C:
/Users/thala/Downloads/FODS/21ST.py --------------------------------------------
========================
Enter sample size: 10
Enter confidence level (e.g. 0.95): 0.95
Enter desired precision: 0.5
Sample Mean: 12.74
Confidence Interval: (np.float64(12.35), np.float64(13.13))
Desired precision achieved
>>>
================ RESTART: C:/Users/thala/Downloads/FODS/22ND.py ================
Average Rating: 4.2
95% Confidence Interval: (np.float64(3.64), np.float64(4.76))
Customer Satisfaction: High
>>>
```

Ln: 16 Col: 0

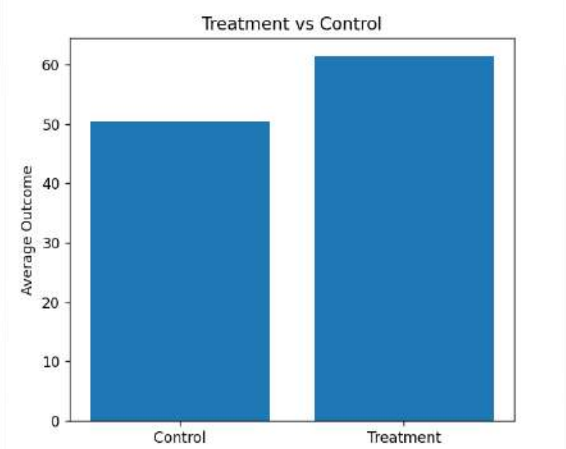
23RD.py - C:/Users/thala/Downloads/FODS/23RD.py (3.14.3)

```python
import numpy as np
from scipy.stats import ttest_ind
import matplotlib.pyplot as plt

control = [50, 52, 49, 51, 50, 48, 52, 51]
treatment = [60, 62, 59, 61, 63, 60, 64, 62]

t_stat, p_value = ttest_ind(control, treatment)

print("T-statistic:", round(t_stat, 2))
print("P-value:", p_value)

if p_value < 0.05:
    print("Treatment has a statistically significant effect"
else:
    print("Treatment has no statistically significant effect

plt.bar(["Control", "Treatment"],
        [np.mean(control), np.mean(treatment)])

plt.title("Treatment vs Control")
plt.ylabel("Average Outcome")
plt.show()
```

*IDLE Shell 3.14.3*

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==================================================================== RESTART: C:
/Users/thala/Downloads/FODS/21ST.py ===========================================
========================
Enter sample size: 10
Enter confidence level (e.g. 0.95): 0.95
Enter desired precision: 0.5
Sample Mean: 12.74
Confidence Interval: (np.float64(12.35), np.float64(13.13))
Desired precision achieved
>>>
================ RESTART: C:/Users/thala/Downloads/FODS/22ND.py ================
Average Rating: 4.2
95% Confidence Interval: (np.float64(3.64), np.float64(4.76))
Customer Satisfaction: High
>>>
================ RESTART: C:/Users/thala/Downloads/FODS/23RD.py ================
T-statistic: -14.17
P-value: 1.0788603492346177e-09
Treatment has a statistically significant effect
```

Figure 1



24TH.py - C:/Users/thala/Downloads/FODS/24TH.py (3.14.3)

File Edit Format Run Options Window Help

```python
from sklearn.neighbors import KNeighborsClassifier

# Training data
X = [
    [1, 2],
    [2, 3],
    [3, 3],
    [6, 7],
    [7, 8],
    [8, 9]
]

y = [0, 0, 0, 1, 1, 1]

# User input
feature1 = float(input("Enter first feature: "))
feature2 = float(input("Enter second feature: "))
k = int(input("Enter value of k: "))

# Create and train model
model = KNeighborsClassifier(n_neighbors=k)
model.fit(X, y)

prediction = model.predict([[feature1, feature2]])

if prediction[0] == 1:
    print("Prediction: Patient has the medical condition")
else:
    print("Prediction: Patient does not have the medical condition")
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
==================================================================== RESTART: C:
/Users/thala/Downloads/FODS/21ST.py ============================================
========================
Enter sample size: 10
Enter confidence level (e.g. 0.95): 0.95
Enter desired precision: 0.5
Sample Mean: 12.74
Confidence Interval: (np.float64(12.35), np.float64(13.13))
Desired precision achieved
>>>
================ RESTART: C:/Users/thala/Downloads/FODS/22ND.py ================
Average Rating: 4.2
95% Confidence Interval: (np.float64(3.64), np.float64(4.76))
Customer Satisfaction: High
>>>
================ RESTART: C:/Users/thala/Downloads/FODS/23RD.py ================
T-statistic: -14.17
P-value: 1.0788603492346177e-09
Treatment has a statistically significant effect
>>>
================ RESTART: C:/Users/thala/Downloads/FODS/24TH.py ================
Enter first feature: 7
Enter second feature: 8
Enter value of k: 3
Prediction: Patient has the medical condition
>>>
```

Ln: 27 Col: 0

25TH.py - C:/Users/thala/Downloads/FODS/25TH.py (3.14.3)

File Edit Format Run Options Window Help

```python
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier

# Load dataset
iris = load_iris()

X = iris.data
y = iris.target

# Train model
model = DecisionTreeClassifier()
model.fit(X, y)

# User input
sepal_length = float(input("Enter sepal length: "))
sepal_width = float(input("Enter sepal width: "))
petal_length = float(input("Enter petal length: "))
petal_width = float(input("Enter petal width: "))

# Prediction
prediction = model.predict([[
    sepal_length,
    sepal_width,
    petal_length,
    petal_width
]])

print("Predicted Flower:", iris.target_names[prediction[0]])
```

IDLE Shell 3.14.3

File Edit Shell Debug Options Window Help

```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> 
================================================================= RESTART: C:
/Users/thala/Downloads/FODS/21ST.py ==========================================
========================
Enter sample size: 10
Enter confidence level (e.g. 0.95): 0.95
Enter desired precision: 0.5
Sample Mean: 12.74
Confidence Interval: (np.float64(12.35), np.float64(13.13))
Desired precision achieved
>>> 
=============== RESTART: C:/Users/thala/Downloads/FODS/22ND.py ===============
Average Rating: 4.2
95% Confidence Interval: (np.float64(3.64), np.float64(4.76))
Customer Satisfaction: High
>>> 
=============== RESTART: C:/Users/thala/Downloads/FODS/23RD.py ===============
T-statistic: -14.17
P-value: 1.0788603492346177e-09
Treatment has a statistically significant effect
>>> 
=============== RESTART: C:/Users/thala/Downloads/FODS/24TH.py ===============
Enter first feature: 7
Enter second feature: 8
Enter value of k: 3
Prediction: Patient has the medical condition
>>> 
=============== RESTART: C:/Users/thala/Downloads/FODS/25TH.py ===============
Enter sepal length: 5.1
Enter sepal width: 3.5
Enter petal length: 1.4
Enter petal width: 0.2
Predicted Flower: setosa
>>> 
```

Ln: 34 Col: 0